

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
(Deemed to be University)
M.Tech /Ph.D END SEMESTER EXAMINATIONS : Nov. / Dec. 2022

Course (Subject) : Computer Oriented Optimization Techniques Code AM-603-D
Duration : 03 hrs Max Marks : 50

Note: 1 Before attempting, please ensure that you have the correct question paper.

2. Attempt **all Five** questions

3. Students may request the invigilator for required Tables (if any).

1). Solve the following L.P.P. by Simplex method (10 M)

$$\begin{aligned} \text{Maximize } Z &= -x_1 + 3x_2 - 2x_3 \\ \text{S.to. } 3x_1 - x_2 + 2x_3 &\leq 7 \\ -2x_1 + 4x_2 &\leq 12 \\ -4x_1 + 3x_2 + 8x_3 &\leq 10 \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

2). Using Kuhn-Tucker conditions to solve the following NLPP. (10 M)

$$\begin{aligned} \text{Max } Z &= -x_1^2 - x_2^2 - x_3^2 + 4x_1 + 6x_2 \\ \text{S.to. } x_1 + x_2 &\leq 2 \\ 2x_1 + 3x_2 &\leq 12 \\ x_1, x_2 &\geq 0 \end{aligned}$$

3. a). Find the extreme points and their nature of the function (5+5 = 10 M)

$$f(x_1, x_2) = x_1^3 + x_2^3 + 2x_1^2 + 4x_2^2 + 6$$

b). Write a short note on Genetic Algorithm and its applications.

4. Use Steepest Decent Method to solve the following problem. (10 M)

$$\text{Minimize } f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2 \text{ starting from the point } X_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

5. A small project is composed of seven activities whose time estimates are listed in the table as follows: (10 M)

Activity		1-2	1-3	1-4	2-5	3-5	4-6	5-6
time estimates in weeks	t_o	5	1	2	3	1	2	1
	t_m	6	1	4	6	1	2	4
	t_p	7	7	12	15	1	8	7

a). Draw the Project net work

b). Find the critical path of the project and project duration.

c). What is the probability that the project will be completed at least 4 weeks earlier than expected. duration.

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